

CONTROL

QWEN2.5-7B-INSTRUCT

— Boundary token → next sentence, R^2 (WikiText, mean of 4 tokens, 1,200 pairs)

PREDICTOR

METRIC

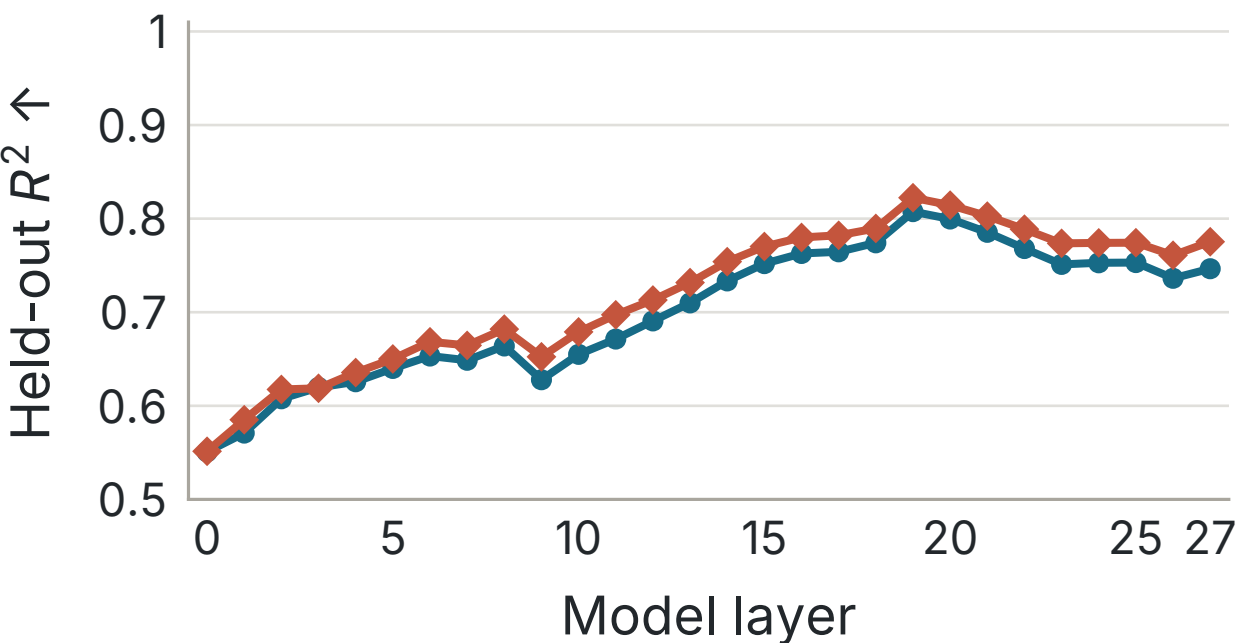
—●— Linear —◆— Nonlinear

A · 50,000 TRAINING CONTEXTS

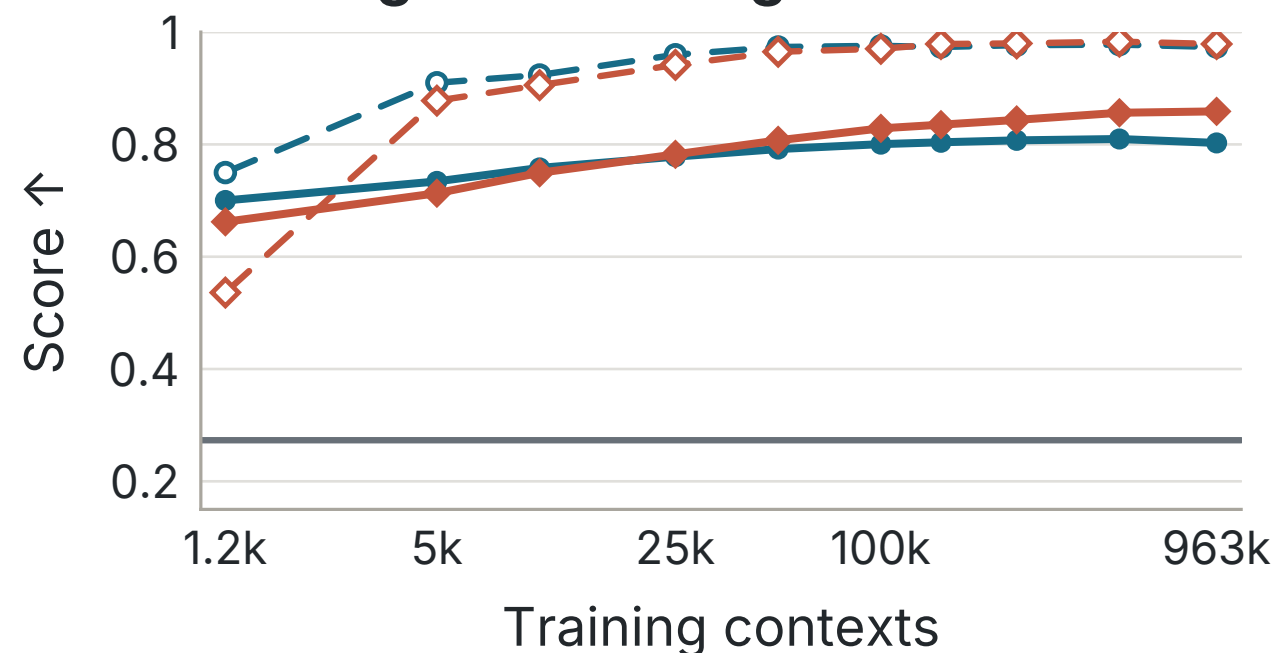
—●— Held-out R^2 —○— Top-1 retrieval

B · LAYER 19

Predictability across layers



Scaling with training data

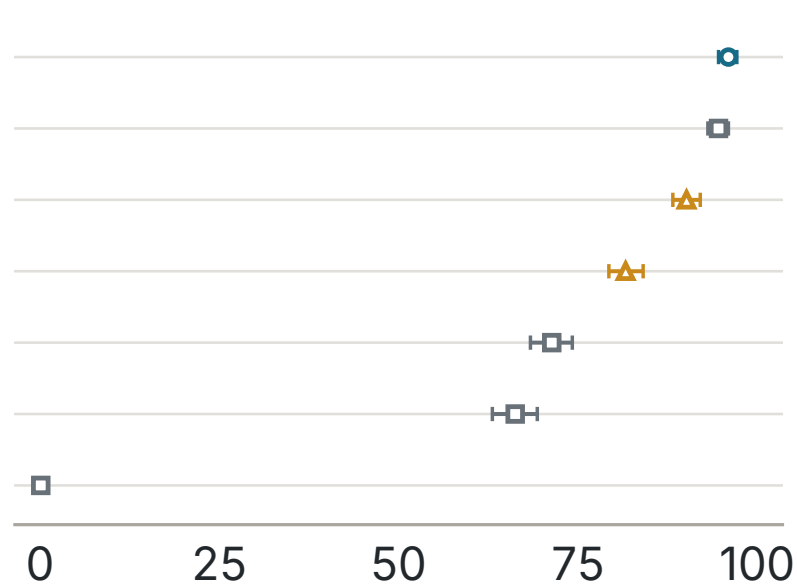
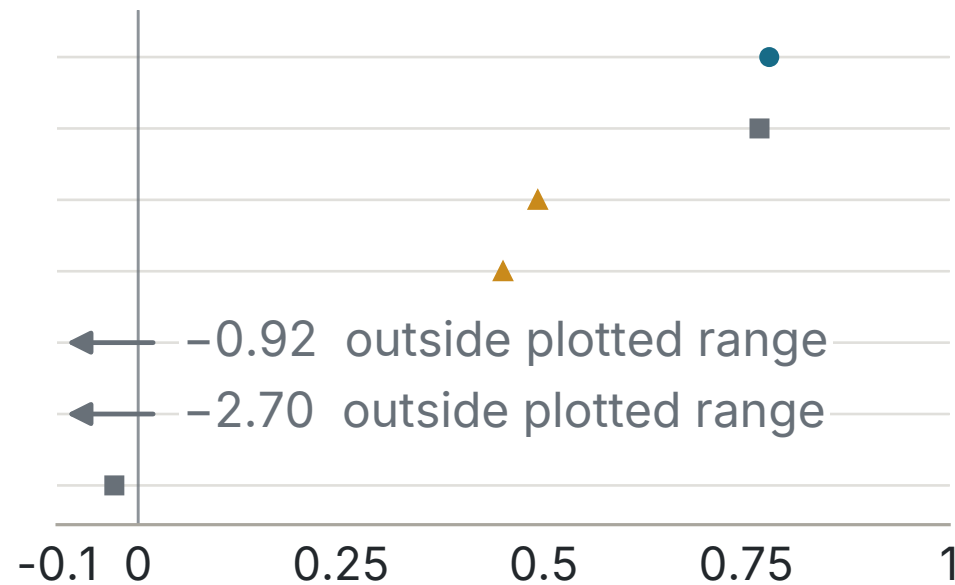


C · LAYER 19, 25,000 CONTEXTS

Baselines

Top-1 retrieval →

Linear map
PCA-1024
Encoder (e5)
Encoder (BGE)
Copy + bias
Copy
Shuffled pairs

Held-out R^2 →

← -0.92 outside plotted range

← -2.70 outside plotted range

Accuracy (%)